TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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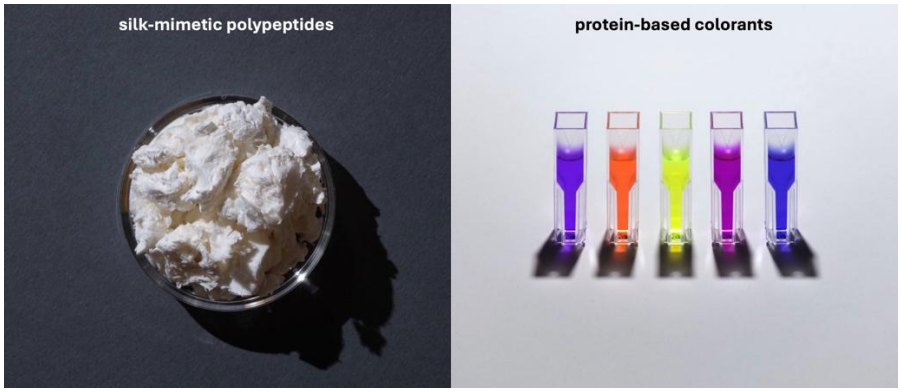
2. Technology Overview

Title of Proposed Technology	**Recombinant Production of Silk Polypeptides and Protein-Based Colorants for Cosmetics and Personal Care Applications**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Industrial Biotechnology	Bio-cosmetics / Materials	Bio-Processes / Bio-Equipment	Bio-engineering Technologies
Concept of Proposed Technology	**Target Industry and Market Need:** The cosmetics and personal care industries have traditionally relied on petroleum-based chemicals, polymers, and colorants. However, health-conscious consumers increasingly demand natural or biobased ingredients made by sustainable, animal-free manufacturing methods. Viscosity modifiers, film-forming agents, humectants, and colorants are major ingredient categories that, to date, require further development of biobased alternatives. The use of synthetic polymers, such as polyethylene glycol, raises concerns about possible immunogenicity with long-term use. Certain azo dyes have been shown to be converted into carcinogens by skin bacteria. Current and potential future regulations, alongside consumer sentiment, drive the need for biobased alternatives. **Technology Platform:** Collaborative efforts between the Zha and Koffas Labs in RPI's Department of Chemical and Biological Engineering have developed multiple synthetic biology platforms for			

	creating bespoke high-performance protein-based ingredients for cosmetics and personal applications. These platforms include recombinant strains of industrially relevant bacteria engineered to produce a panel of silk-mimetic polypeptides[1-5] that can be used as humectants, viscosity modifiers, and film-forming agents. Compared to commercially available silkworm-derived silk protein, our silk-mimetic polypeptides can be readily tuned for better performance in formulations or for enhanced bioactivity. Moreover, we have engineered host strains for more optimal performance in upstream and downstream bioprocessing of silk-mimetic polypeptides. Additionally, we have engineered microbial platforms for producing coral-inspired protein colorants for use in cosmetics and personal care products. Our protein colorants span a palette of purple, pink, orange, blue, and green. Unlike azo dyes or natural pigments, these proteins can be mutated to display different colors. Moreover, as they are based on natural amino acids, these protein colorants are naturally biodegradable and biocompatible.  **<u>Academic Team:</u>** Prof. Helen Zha is an associate professor and Dean's Faculty Fellow in the Department of Chemical and Biological Engineering. Her research expertise encompasses 18 years of experience in the synthesis and study of biomimetic materials, *de novo* designed peptide and protein-based materials, and self-assembled soft matter. Prof. Mattheos Koffas is the Dorothy and Fred Chau '71 Endowed Professor in the Department of Chemical and Biological Engineering. His research expertise lies in synthetic biology and metabolic engineering, including developing recombinant production platforms for plant natural products, animal polysaccharides such as chondroitin sulfate and spider silk protein, and polyamino acids.
Rationale for Joint R&D Project	We are seeking industry collaborators with interest in cosmetics, personal care, and/or biomedical ingredients made by advanced fermentation to co-develop and optimize our platforms. Co-development in collaboration with an industry partner will help provide concrete, impactful performance and price targets. Additionally, industry partnership will provide the possibility of evaluation by potential customers.
Key Words	1. Synthetic biology 2. Silk-mimetic polypeptides 3. Protein colorants 4. Advanced fermentation 5. Cosmetics and personal care ingredients

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (7)
R&D Objectives	The overall objective of this project is to develop recombinant silk-mimetic polypeptides and protein-based colorants for use in cosmetics and/or personal care formulations that enhance the performance and value of these products. Specific goals include (1) engineering protein sequences to improve processing efficiency or performance in conditions simulating real-world use, (2) creating recombinant host strains that produce these proteins with greater yield, and (3) optimizing fermentation and purification processes for quality and cost efficiency.			
R&D Target Specifications	Target specifications will be developed in collaboration with an industry partner. Specifications may include physicochemical attributes, such as molecular weight, solubility, or color. Specifications may also include performance attributes (such as stability in combination with other ingredients or in usage/storage environments), other critical quality attributes (such as purity), or manufacturing productivity (such as fermentation yield or purification costs).			

R&D Plan	The Research Plan, including milestones, will be solidified in collaboration with the industry partner to best meet target specifications (described above) and desired milestones. Research tasks with approximate timeline can include: **Stage 1: Protein Engineering (month 0–12):** • Test our existing panel of silk-mimetic polypeptides and/or protein colorants for performance (according to target specifications) and benchmarking against existing solutions • Re-design protein sequences to improve performance or manufacturability • Iterate a design-build-test-learn process to approach target specifications • If possible or desirable, obtain customer feedback to guide development **Stage 2: Fermentation Optimization and Scale-Up (month 12–18):** • Optimize parameters for fed-batch fermentation at scales up to 30 L • Engineer or select new host strains for improved fermentation yields, if necessary • Perform technoeconomic analysis to guide optimization **Stage 3: Downstream Processing Optimization (month 18–24):** • Optimize the downstream bioprocessing workflow, including removing host cell contaminants, isolating target proteins, and processing for storage or further integration into a cosmetic or personal care product • Perform technoeconomic analysis to guide process optimization

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The RPI research team will lead and perform technical tasks as described in the R&D plan related to designing, synthesizing, and purifying recombinant proteins at bench and pilot (up to 30 L) scales. Major equipment (e.g., 30 L fermenter) for working with and engineering recombinant microbes is available through the Microbiology and Fermentation Core facility and the Genomics Core Facility at RPI. Major equipment for physicochemical characterization of recombinant proteins is available through the Analytical Biochemistry Core facility at RPI. Additional facilities and equipment for recombinant protein production and strain engineering is available in the research labs of Profs. Zha and Koffas. If needed, specialized equipment for protein purification and isolation can be included in the project budget and installed at RPI.
Expected R&R of Korean SME Partner	The Korean SME will collaborate with the RPI research team to define target specifications and determine appropriate benchmarks and milestones. Furthermore, the Korean SME will carry out performance testing of the recombinant proteins and polypeptides synthesized by the RPI team, particularly to ensure that real-life storage, processing, and usage environments are mimicked. If desired, the Korean SME will also seek customer evaluations as part of the R&D process.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected outcomes include eco-friendly, biocompatible silk-mimetic polypeptides and protein-based colorants for use in cosmetics or personal care formulations that enhance the performance and value of these products. Other technical outcomes include optimized host strains, fermentation processes, and purification workflows for producing these polypeptides and colorants. We aim to reachTRL 7 by the end of the project, which will establish solid foundations for product development and manufacturing for future commercialization efforts.

6. References

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5. A. J. Connor, M. Koffas, D. Nelson, **R. H. Zha**. Systems and methods for increased production of recombinant biopolymers via genome engineering and downregulation of basal expression. US12391912B2. Granted August 19, **2025**.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization